1
2
´a
á
$y; v$
y
$v(F_Y)$
$y; v$
$v(F_Y)+$
$IF(y;v)$
$E[RIF(y;v)] =$
$v(F_Y)$
$y; v$

$$E[RIF(Y;v)|X] = X\gamma + \epsilon, \quad (1)$$

γ
Y, Q_τ
$(\tau -$
$I\{Y \leq$
$Q_\tau\})/f_Y(Q_\tau)$
$I\{\cdot\}$
$f_Y(\cdot)$
Y
Q_τ
τ
Y
$Y; Q_\tau$
$Q_\tau +$
$IF(Y, Q_\tau)$

$$RIF(Y;Q_\tau) = Q_\tau + \frac{\tau - I\{Y \leq Q_\tau\}}{f_Y(Q_\tau)} = c_{1,\tau} \cdot I\{Y > Q_\tau\} + c_{2,\tau}, \quad (2)$$

$c_{1,\tau} =$
$\frac{1}{f_Y(Q_\tau)}$
$c_{2,\tau} =$
$Q_\tau -$
$c_{1,\tau} \cdot$
$(1 -$
$\tau)$
$c_{1,\tau}$
$c_{2,\tau}$
$I\{Y \leq$
$Q_\tau\}$

The decomposition approach depends on the order in which the decomposition is performed Fortin2011 Using the CQRs, we can only interpret